SECTION 2 — Overall Description

2.1 User Classes and Characteristics

There is one primary user class for this system:

Students/Job Seekers: Individuals applying for jobs or internships who need to tailor their resume and cover letter to specific postings and track their applications. They are assumed to have basic computer literacy and are comfortable using a web browser, but are not expected to have any technical knowledge of AI or software development.

There is no separate administrator role in the current scope, as the system is designed for individual use rather than institutional management.

2.2 User Needs

Users need a fast way to understand how well their resume matches a specific job.

Users need tailored resume bullet points and cover letters without manually rewriting them for every application.

Users need visibility into which skills they are missing for a target role.

Users need a simple way to track the status of applications they have submitted, without using a separate spreadsheet or notebook.

2.3 Operating Environment

The system will run as a web application accessible through a standard web browser (Chrome, Firefox, Edge).

The backend will be built using FastAPI (Python) and will run on a server capable of hosting a Python web application.

The database will be MySQL, used to store user data, resumes, job postings, and application records.

The AI agent pipeline will be built using Google ADK, which will communicate with an external LLM service (Gemini or Claude) over the internet.

No native mobile application is planned; the system is web-based only.

2.4 Constraints

The system will not perform live scraping of job postings from external websites (e.g., LinkedIn, Indeed) due to reliability issues such as anti-bot protection and frequently changing page structures. Job descriptions must be submitted as pasted text.

The project must be completed within a single academic semester, which limits the scope to core features only.

The system depends on an external LLM provider; any downtime or rate limiting on that provider's side will directly affect system availability.

Development will be carried out by a four-member student team with limited prior experience in multi-agent AI systems.

2.5 Assumptions

It is assumed that users will have a resume ready in PDF format before using the system.

It is assumed that job descriptions will be provided in English.

It is assumed that users have basic internet access and a modern web browser.

It is assumed that the LLM service used through Google ADK will remain accessible and stable throughout the development and demonstration period.

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